

AP Calculus II
Notes 10.2
Parametric Equations

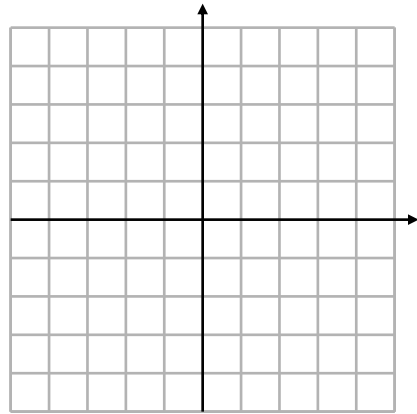
Definition of a Plane Curve: If f and g are continuous functions of t on an interval I , then the equations

$$x = f(t) \quad \text{and} \quad y = g(t)$$

are called **parametric equations** and t is called a **parameter**, usually measuring time. The set of ordered pairs (x, y) obtained as t varies over the interval I is called the **graph** of the parametric equations. Taken together, the parametric equations and the graph are called a **plane curve**, denoted by C . This essentially allows us to determine where the x and y coordinates are over time through the new variable t .

Ex. 1: Sketch the curve denoted by the parametric equations $x = t^2 - 1$ & $y = 2t + 1$.

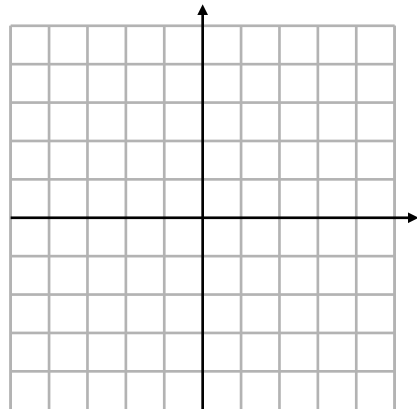
t	x	y
-2		
-1		
0		
1		
2		



The importance of parametric equations is usually found in time and speeds. Here is a good example of how speeds can be determined/manipulated.

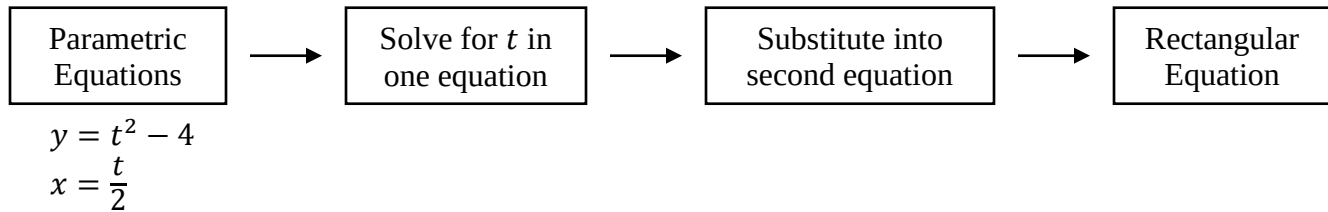
Ex. 2: Sketch the curve denoted by the parametric equations $x = \frac{1}{4}t^2 - 1$ & $y = t + 1$.

t	x	y
-4		
-2		
-1		
0		
1		
2		
4		



Eliminating the Parameter

Finding a rectangular equation that represents the graph of a set of parametric equations is called **eliminating the parameter**. Sometimes it is desirable to eliminate the parameter to recognize the curve.

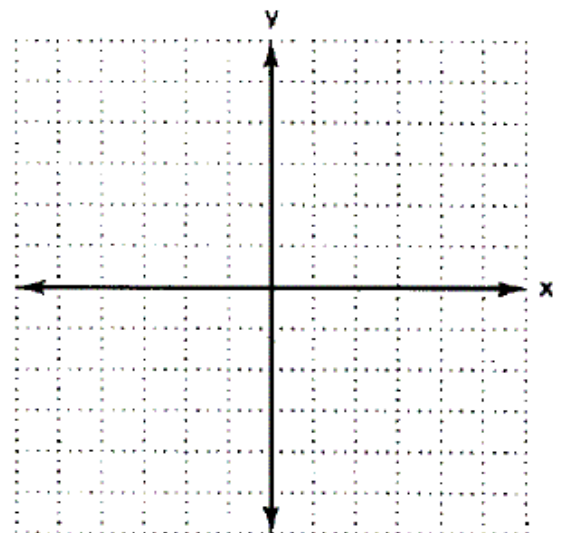


Ex. 3: Rewrite the following equation in parametric form into the rectangular form.

$$x = \frac{1}{\sqrt{t+1}} \text{ and } y = \frac{t}{t+1}, \quad t > -1$$

Ex. 4: Rewrite the following equation in parametric form into the rectangular form and then sketch.

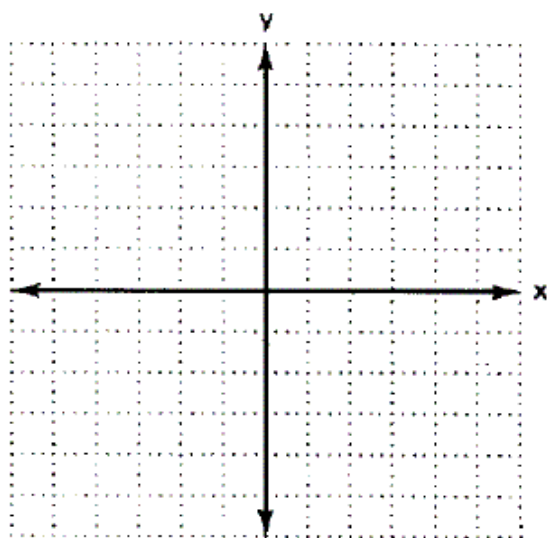
$$x = 5 \sin \theta \text{ and } y = 5 \cos \theta, \quad 0 \leq \theta \leq 2\pi$$



Ex. 5: Sketch the following parametric equation
(including orientation and labeling)
from $-3 \leq t \leq 3$:

$$x = \frac{1}{2}t^3 - 3t$$

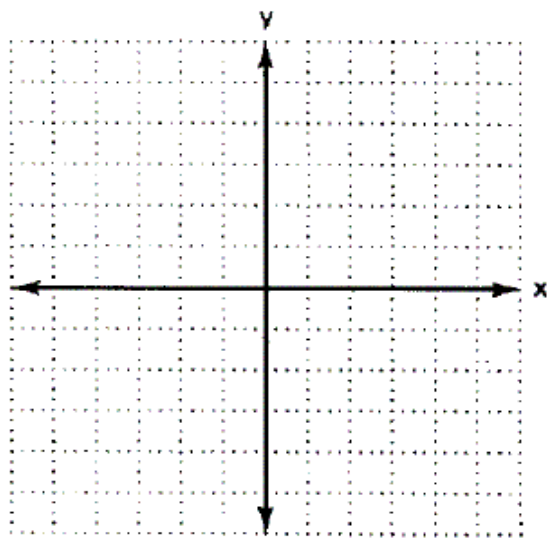
$$y = t^2 - 4$$



Ex. 6: Sketch the following parametric equation
(including orientation and labeling)

$$x = 1 + 3 \cos \theta$$

$$y = -2 + \sin \theta$$



Ex. 7: The left field fence at Camden Yards is 337 feet from home plate and is 7 feet high. Chris Davis hits the baseball at a point 3 feet about the ground. It leaves the bat at an angle of 41° and at a speed of 107 feet per second.

Given $x = (v_0 \cos \theta)t$ $y = h + (v_0 \sin \theta)t - 16t^2$

- a) Write a set of parametric equations for the path of the baseball.

- b) How high is the baseball after 2 seconds in flight?

- c) How long will it take for the ball to reach its maximum height?

- d) Did Chris Davis hit a homerun? Be sure to explain/justify your answer.

AP Calculus II
Notes 10.3
Parametric Equations and Calculus

The slope of the line tangent to a curve described by $x = f(t)$ and $y = g(t)$ is the first derivative of y with respect to x . However, both x and y are in terms of t , so the slope of the tangent line is defined as

$$\frac{dy}{dx} =$$

The parametric curve will have a horizontal tangent when _____

The parametric curve will have a vertical tangent when _____

Ex. 1: For the curve $x(t) = t^3$, $y(t) = t^2 - 2t$

a) Find the equation of the tangent line at the ordered pair (8,0).

b) Find the points of the horizontal and vertical tangents to the curve.

Second Derivative and Concavity

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left[\frac{dy}{dx} \right] = \frac{d/dt}{dx/dt} \left[\frac{dy}{dx} \right] =$$

Ex. 2: Consider the parametric equation $x(t) = 3 + \sin \pi t$, $y(t) = 1 + e^{2t}$.

a) Find $\frac{dx}{dt}$, $\frac{dy}{dt}$ and $\frac{dy}{dx}$ at $t = 1.6$. Then, explain the graphical meaning of each.

b) Determine if the equation is increasing or decreasing and concave up or down at $t = 1.6$.

Ex. 3: Find the first two derivatives of the function determined by $x = 5 \cos \theta$, $y = 4 \sin \theta - 3$ at $\theta = \frac{\pi}{6}$.
Then, find if the curve is getting closer or further away from the x -axis and y -axis at this point.

Arc Length in Parametric

Since both $x(t)$ and $y(t)$ are both functions with respect to t , the proof of arc length in rectangular form is expanded to convert it to parametric equations:

Ex. 4: Find the length of the curve whose parametric equation is $x = \cos^3 t$ and $y = \sin^3 t$ from $\left[0, \frac{\pi}{2}\right]$.

PVA – Related Definitions in Parametric/Vector Form

Suppose a particle moves along a smooth curve in the plane so that its position at any time t is $\langle x(t), y(t) \rangle$ where x and y are both twice-differentiable functions of t .

- 1) The particle's **position** -
- 2) The particle's **velocity** -
- 3) The particle's **acceleration** -
- 4) The particle's **speed** -
- 5) The particle's **distance travelled** –
- 6) The particle's **displacement** –
- 7) The particle's **at rest when** –

A graphing calculator is required for problems on this part of the exam.

At time $t \geq 0$, the position of a particle moving along a curve in the xy -plane is $(x(t), y(t))$, where $\frac{dx}{dt} = t - 5 \cos t$ and $\frac{dy}{dt} = 6 \cos(1 + \sin t)$. At time $t = 3$, the particle is at position $(-1, 2)$.

- (a) Write an equation for the line tangent to the path of the particle at time $t = 3$.
- (b) Find the time t when the line tangent to the path of the particle is vertical. Is the direction of motion of the particle up or down at that moment? Give a reason for your answer.
- (c) Find the y -coordinate of the particle's position at time $t = 0$.
- (d) Find the total distance traveled by the particle for $0 \leq t \leq 3$.

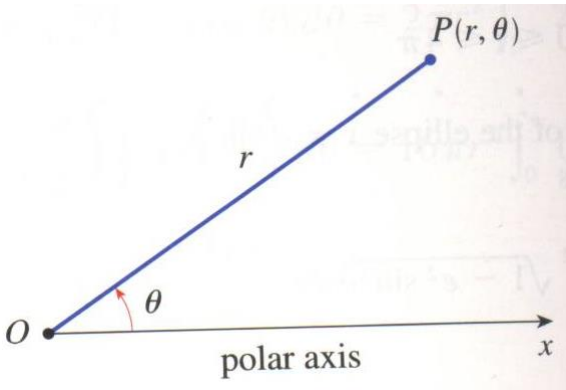
a) Find the position of the particle at time $t = 0$.

b) Find the time, if any, when the particle is at rest.

c) Find the acceleration vector at time $t = \frac{1}{2}$.

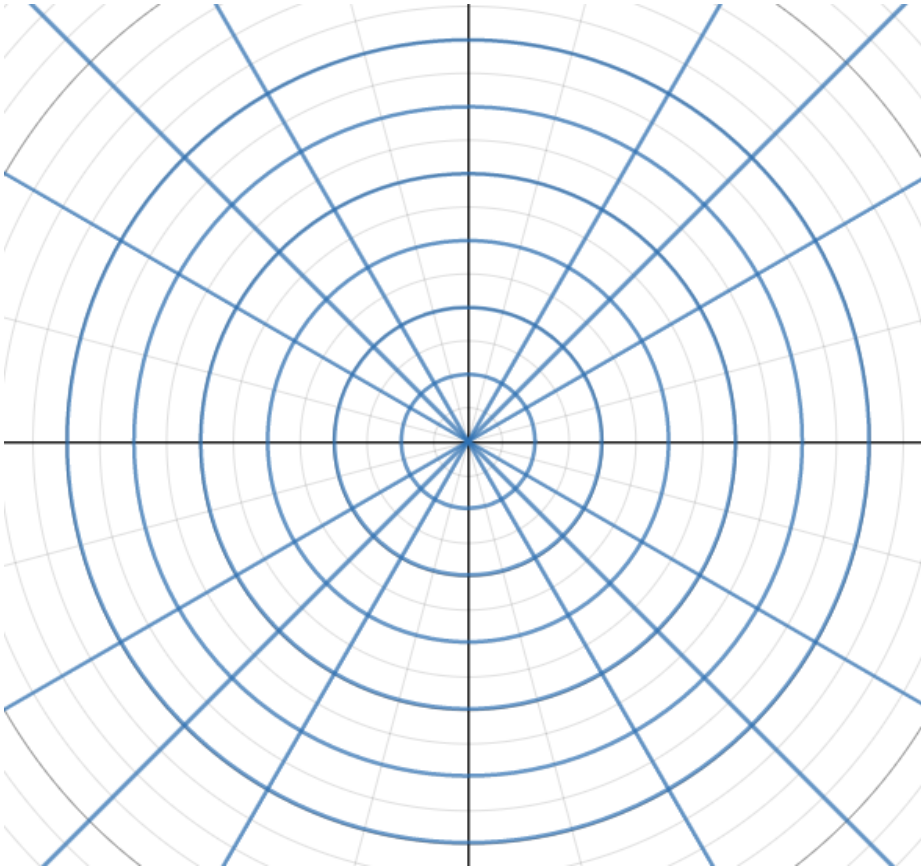
AP Calculus II
Notes 10.4
Polar Coordinates

So far, all the graphs you have been working with since middle school are in the rectangular form with points (x, y) . Plotting points always required you to move left or right and then up or down. A new coordinate system is called the **polar coordinate system**.



To form a point in this system, fix a point O , called the **pole** or **origin**, and a ray emanating from O , called the **polar axis**. To locate a point P , in the plane, you draw a segment OP , r is the length of OP and θ is the angle from the polar axis to OP measured in the counterclockwise direction. Polar coordinates are written as (r, θ) .

Graphing Polar Coordinates Practice:



- a) $\left(2, \frac{\pi}{3}\right)$
- b) $\left(4, \frac{5\pi}{6}\right)$
- c) $\left(5, \frac{3\pi}{2}\right)$
- d) $\left(1, \frac{3\pi}{4}\right)$
- e) $(1.5, 0)$
- f) $\left(0, \frac{4\pi}{3}\right)$
- g) $\left(-2, \frac{\pi}{6}\right)$
- h) $\left(3, -\frac{\pi}{3}\right)$
- i) $\left(-4, \frac{11\pi}{6}\right)$

Rectangular $\longleftrightarrow$ Polar Conversion:

Ex. 1: Convert the following polar coordinates into rectangular coordinates:

a) $(r, \theta) = (2, \pi)$

b) $(r, \theta) = \left(\sqrt{3}, \frac{\pi}{6}\right)$

Ex. 2: Convert the following rectangular coordinates into 2 sets of polar coordinates:

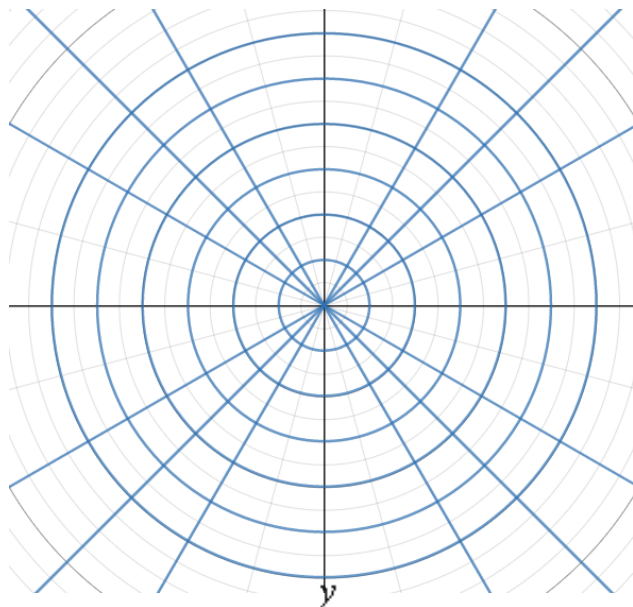
a) $(x, y) = (-1, 1)$

b) $(x, y) = (0, 4)$

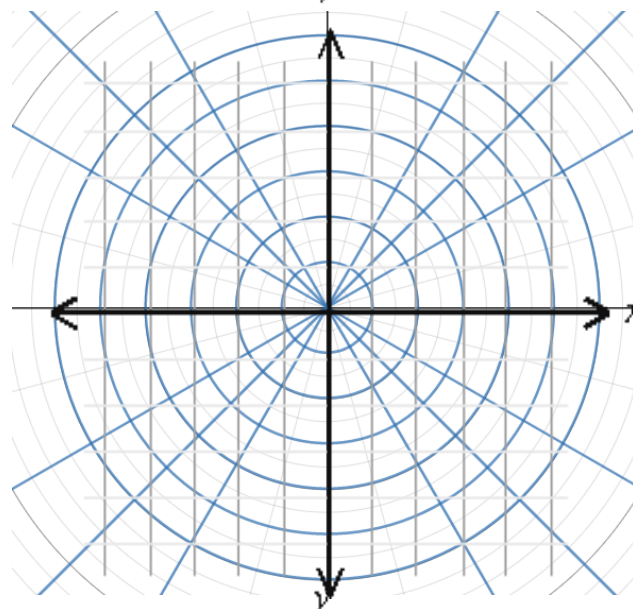
Graphing Polar Equations

Ex. 3: Graph the following polar equations:

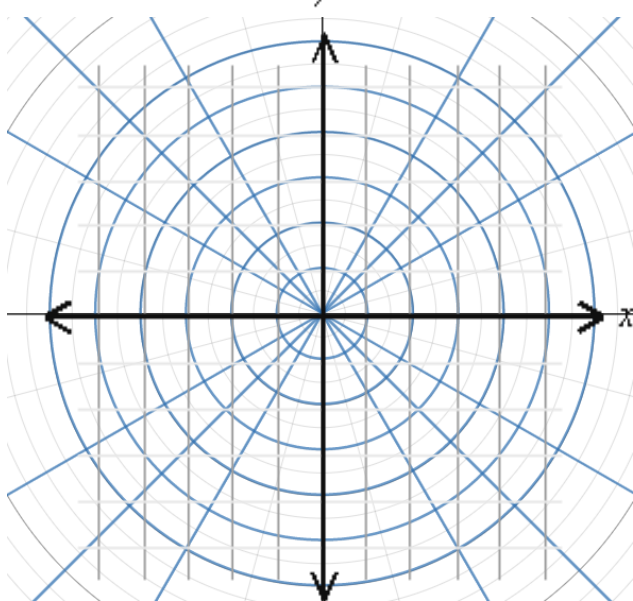
a) $r = 2$



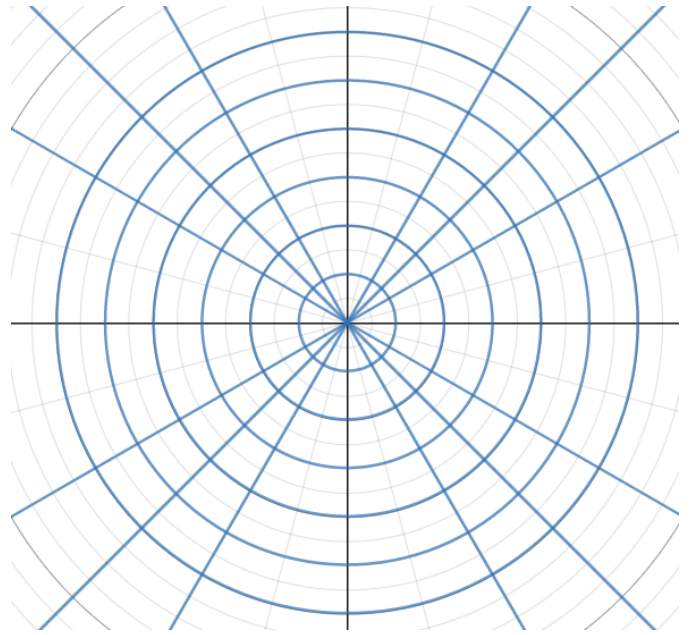
b) $\theta = \frac{\pi}{3}$



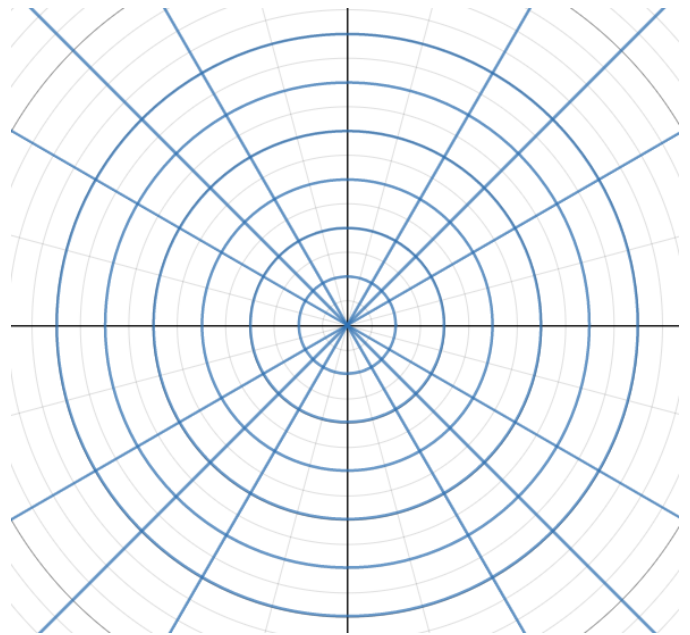
c) $r = 2 \csc \theta$



d) Limaçon: $r = 2 - 4 \sin \theta$



e) Rose: $r = 4 \cos 3\theta$

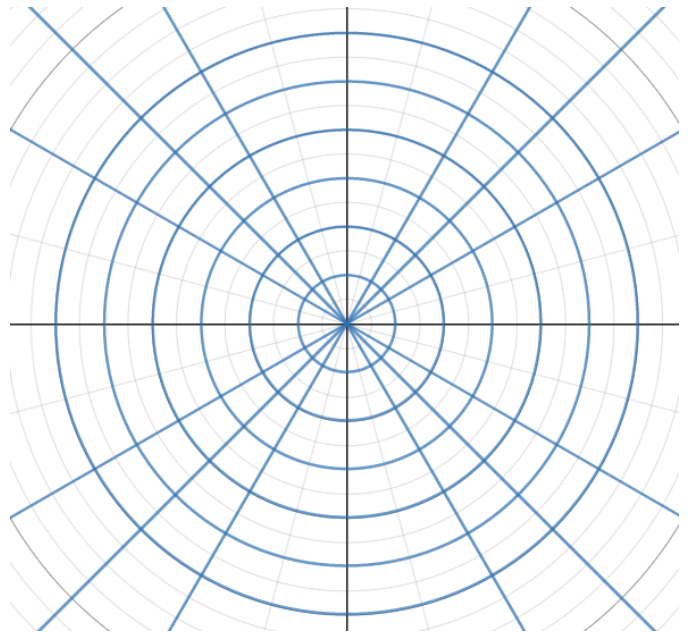


Polar Coordinates and Calculus

If f is a differentiable function of θ , then the *slope* of the tangent line to the graph of $r = f(\theta)$ at the point (r, θ) is found the same way as a parametric equation. In fact, we will convert polar equations to parametric equations using the relationships:

$$\frac{dy}{dx} =$$

Ex. 4: Find $\frac{dr}{d\theta}$, $\frac{dy}{d\theta}$, $\frac{dx}{d\theta}$ and the slope of the tangent line to $r = 4 \cos \theta$ at $\theta = \frac{\pi}{4}$ and then, explain the meaning of each.



Horizontal tangent lines occur when –

Vertical tangent lines occur when –

Ex. 5: Find all points (r, θ) of horizontal and vertical tangency to $r = 2(1 - \cos \theta)$ from $0 \leq \theta < 2\pi$.

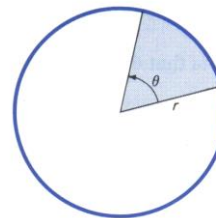
Ex. 6: Find the equation of the tangent line to $r = \sin \theta$ at $\theta = \frac{\pi}{3}$.

AP Calculus II
Notes 10.5
Polar Area and Arc Length

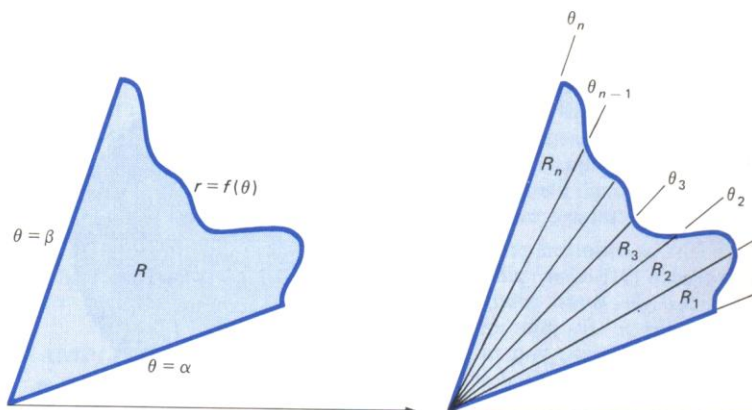
The development of the polar area formula is similar to that in the rectangular coordinate system. Instead of using rectangles, this system uses sectors of a circle as the basic elements that are being added up.

Proof Using Sectors of a Circle –

In Geometry, we learned that the area of a sector of a circle is $\frac{n^\circ}{360^\circ}(\pi r^2)$. Since we use radians in Calculus, we will translate that to where θ is the fraction of the circle or angle and $r = f(\theta)$.



Just like in rectangular mode, where area is infinitely many rectangles added together, polar area has infinitely many sectors added together. So, the area of a region bounded by $\theta = a$, $\theta = b$ and $r = f(\theta)$ where f is continuous and positive and $a \leq \theta \leq b$ is...



Area of a Region in Polar

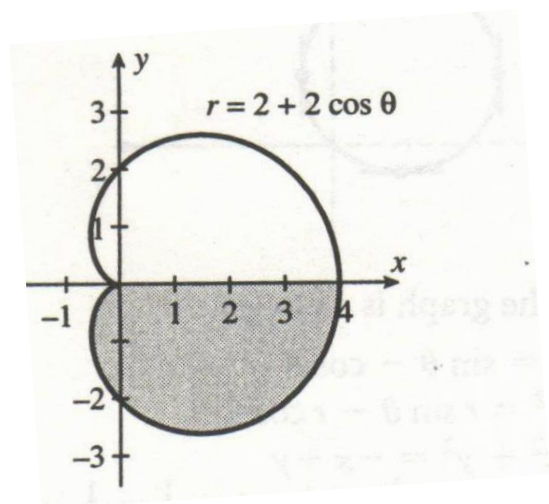
Ex. 1: Find the area enclosed by $r = 2 \sin \theta$.

Identities you MUST know:

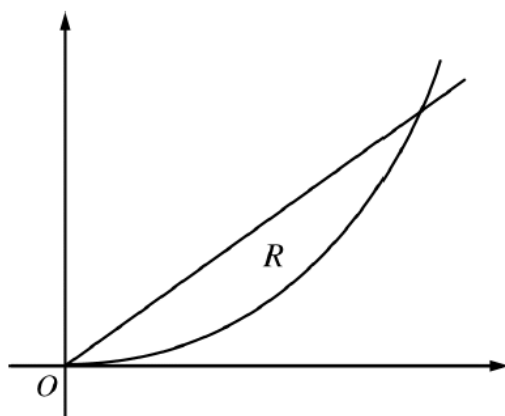
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Ex. 2: Find the area of $r = 2 + 2 \cos \theta$ below the polar axis, as shown in the figure.

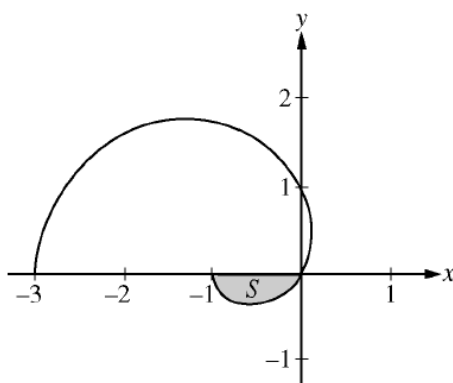


Ex. 3:



Let R be the region in the first quadrant that is bounded by the polar curves $r = \theta$ and $\theta = k$, where k is a constant, $0 \leq k \leq \frac{\pi}{2}$, as shown in the figure. What is the area of R in terms of k ?

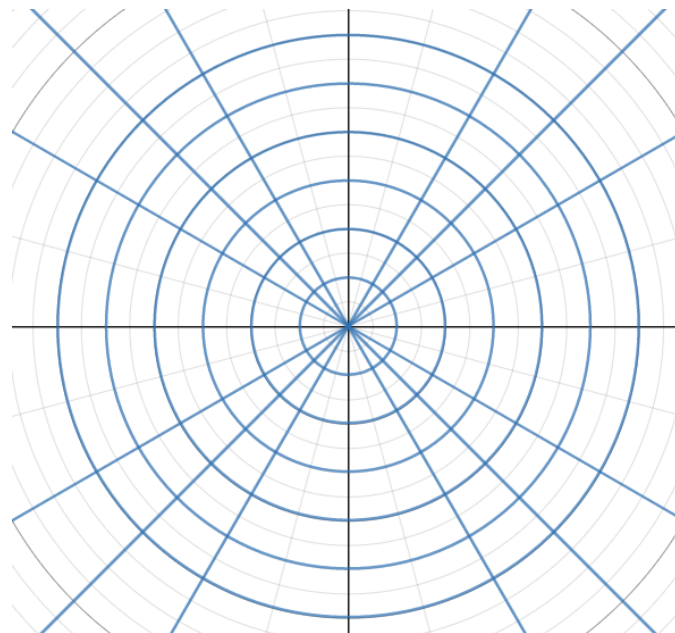
Ex. 4:



The graph of the polar curve $r = 1 - 2 \cos \theta$ for $0 \leq \theta \leq \pi$ is shown above. Let S be the shaded region in the third quadrant bounded by the curve and the x -axis.

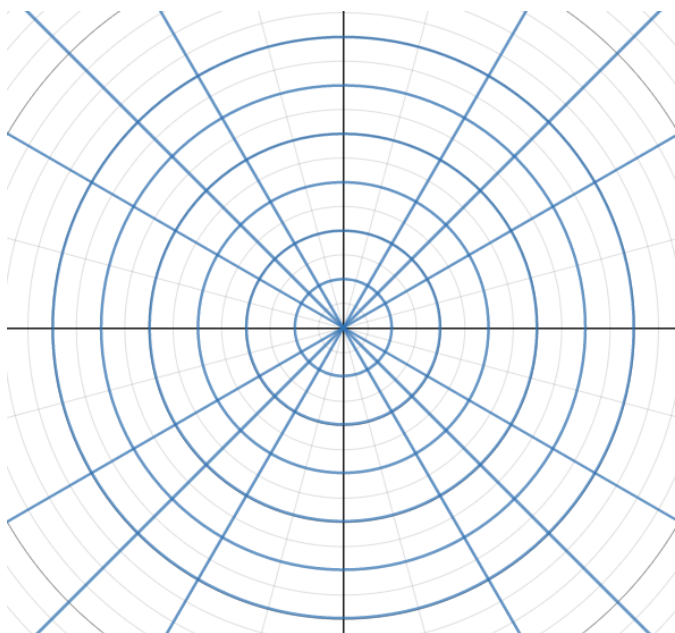
- Write an integral expression for the area of S .
- Write expression for $\frac{dx}{d\theta}$ and $\frac{dy}{d\theta}$ in terms of θ .
- Write an equation in terms of x and y for the line tangent to the graph of the polar curve at the point where $\theta = \frac{\pi}{2}$. Show the computations that lead to your answer.

Ex. 5: Find the area in the inner loop of $r = 2 - 4 \sin \theta$.

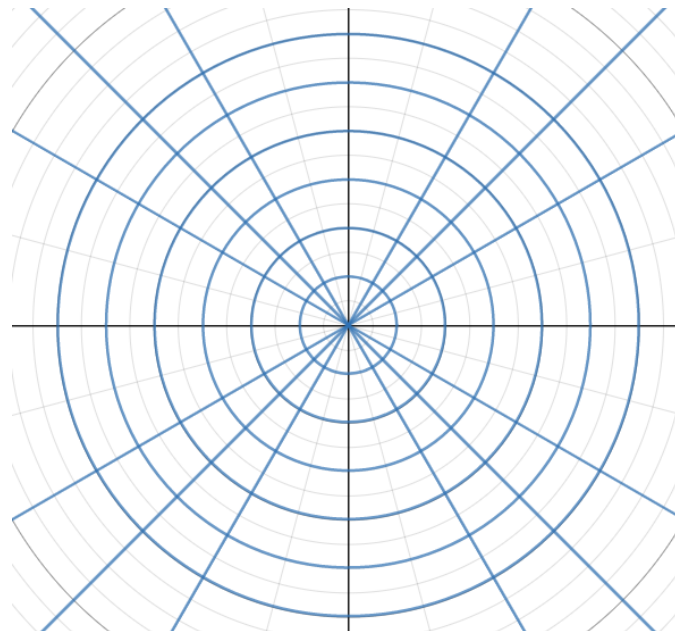


Area Between Two Curves

Ex. 6: Find the area of the region inside the polar curve $r = 4 \sin \theta$ and outside the polar curve $r = 2$.

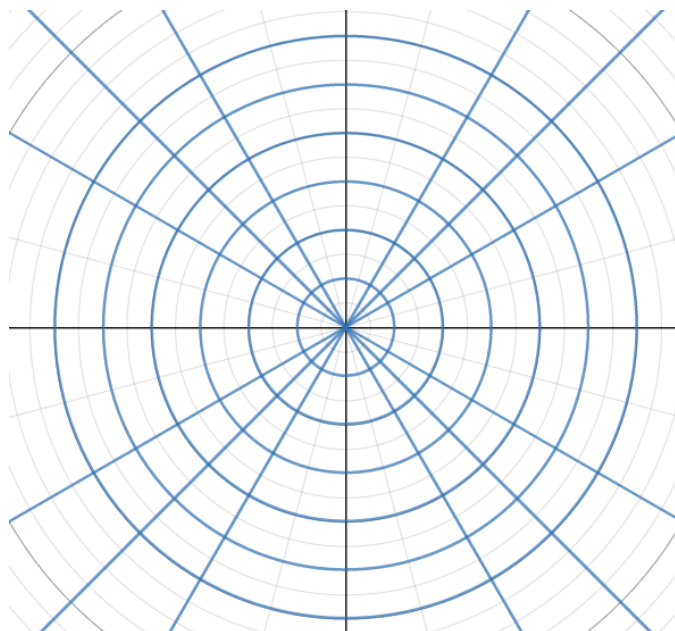


Ex. 7: Find the area of the region inside the curve $r = 4 \cos \theta$ and outside the curve $r = 2 \cos \theta$.

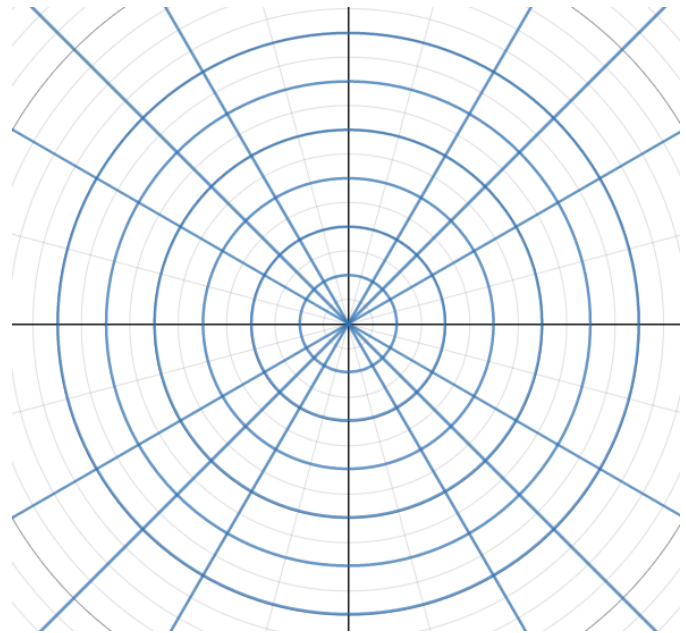


Ex. 8: Setup, but do not evaluate, the area of the region described.

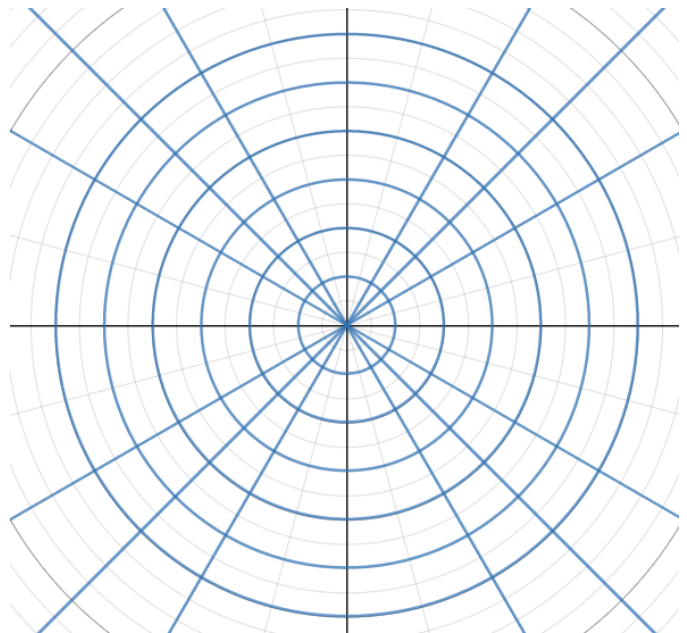
- a) Common interior of $r = 4 \cos \theta$ and $r = 2$.



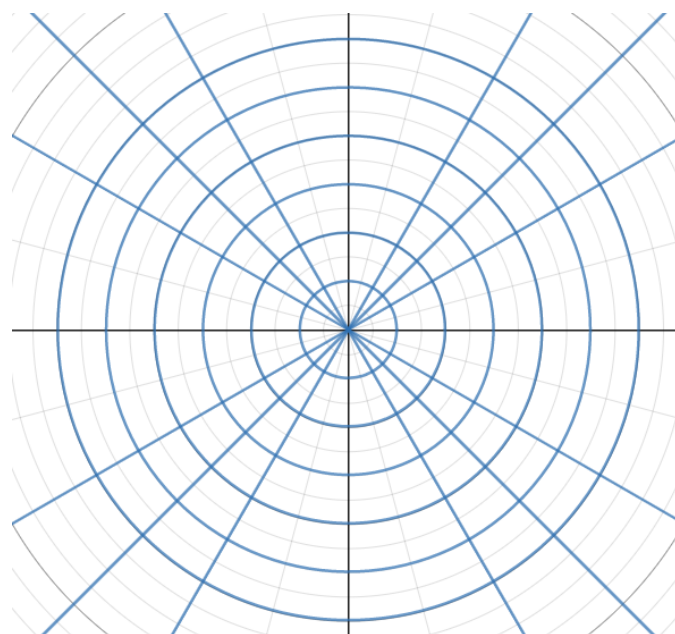
- b) Common interior of $r = 3 \cos \theta$ and $r = 1 + \cos \theta$.



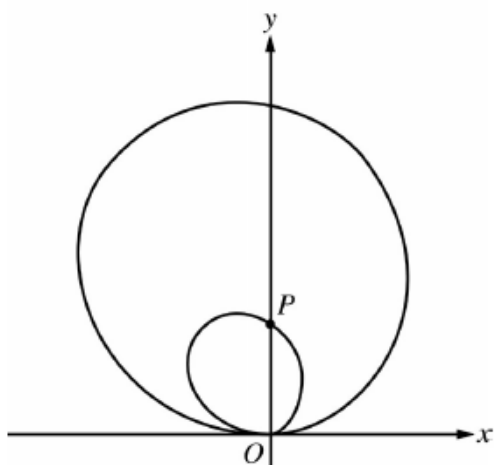
- c) Inside $r = 2$ and outside $r = 2 - 2 \sin \theta$. Then, determine at what rate the distance between the two functions is changing when $\theta = \frac{2\pi}{3}$.



d) Enclosed by $r = 3 \cos \theta$ and $r = 2 - \cos \theta$.



Ex. 9: A graphing calculator is required for problems on this part of the exam.

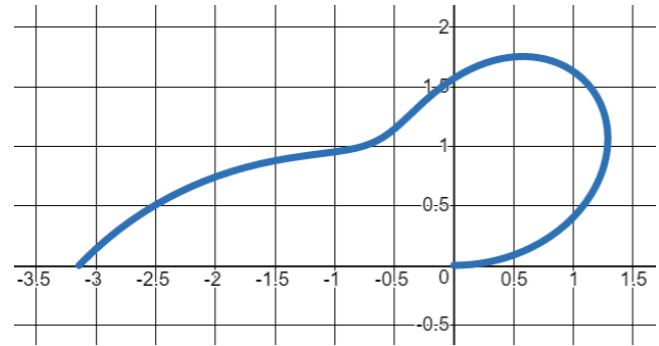


Let r be the function given by $r(\theta) = 3\theta \sin \theta$ for $0 \leq \theta \leq 2\pi$. The graph of r in polar coordinates consists of two loops, as shown in the figure above. Point P is on the graph of r and the y -axis.

- (A) Find the rate of change of the x -coordinate with respect to θ at the point P .
- (B) Find the area of the region between the inner and outer loops of the graph.

Ex. 10: CA The curve shown below is drawn in the xy -plane and is described by the equation in polar coordinates $r = \theta + \sin 2\theta$ for $0 \leq \theta \leq \pi$, where r is measured in meters and θ is in radians.

- a) Find the area bounded by the curve and the x -axis.



- b) Find the angle θ that corresponds to the point on the curve with x -coordinate -2 .

- c) For $\frac{\pi}{3} < \theta < \frac{2\pi}{3}$, $\frac{dr}{d\theta}$ is negative. What does this fact say about r ? What does this fact say about the curve?

- d) Find the value of θ in the interval $0 \leq \theta \leq \frac{\pi}{2}$ that corresponds to the point on the curve in the first quadrant with greatest distance from the origin. Justify your answer.

Arc Length

When we want to find the arc length of a region in polar, we think of this parametrically. The only difference is the instead of t and dt , we will be using θ and $d\theta$. This gives you:

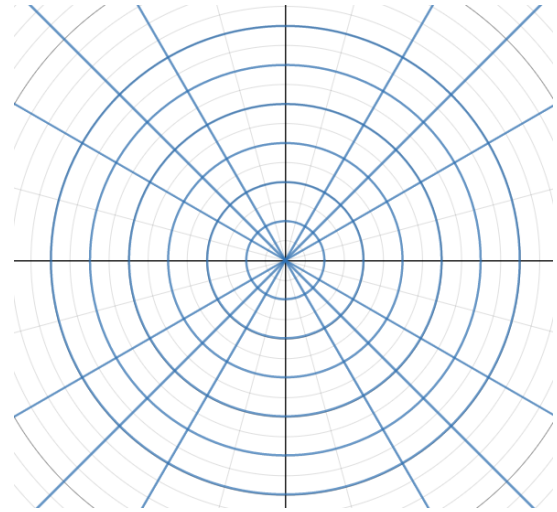
Rectangular:

Parametric:

Polar:

Ex. 11: Find the length of the spiral $r = e^\theta$ between $\theta = 0$ and $\theta = \frac{\pi}{2}$.

Ex. 12: Find the length of the graph of $r = 3 \cos \theta$.



Ex. 13: Find the length of the polar curve $r = 2 \sec \theta$ from $\theta = 0$ to $\theta = \frac{\pi}{3}$.